

Sovereign Debt Sustainability in Developed Economies

A Theoretical Review, Cross-Country Considerations, and Evidence on Wealth Effects in the United States

Rodrigue Girard*

This Version: November 2025

Abstract

This dissertation examines the concept of sovereign debt sustainability in advanced economies, questioning whether universal thresholds or tools can meaningfully capture country-specific realities. It reviews the main theoretical frameworks, from the intertemporal government budget constraint to Bohn's fiscal reaction function and the IMF's debt sustainability analysis framework. It argues that sustainability assessments must go beyond static debt ratios and incorporate the notion of fiscal space to better account for heterogeneity across countries. The dissertation also presents an empirical analysis of the determinants of fiscal space in advanced economies. Using data for the United States, it shows that increases in net private financial wealth are associated with lower long-term government bond yields when public debt is predominantly domestically held.

NB: This is an extract of my Master's thesis with an extended econometric analysis, submitted as part of my MSc in Finance at Université Paris Sciences et Lettres - Paris Dauphine, and which received a grade of 87.5%. The original submitted version comprised more than 13,000 words and can be sent upon request, containing a more rigorous explanation of the framework and a deeper analysis of the results.

*London School of Economics, Université Paris Sciences et Lettres - Paris Dauphine

Contents

Introduction	1
1 Critical Literature Review of the Main Methods to Assess Sustainability	2
1.1 The Traditional IGBC Framework and Its Limitations	2
1.2 Debt Sustainability under a more realistic framework: Bohn (1995, 1998)	3
1.3 Stabilization-Based Approaches: Debt Dynamics and IMF DSA	5
2 Fiscal Space and Country-Specific Debt Limits	6
2.1 Fiscal Space: Motivation and Core Concept	6
2.2 Fiscal Fatigue, Debt Limits, and Market Expectations	7
3 Long-Term Determinants and Constraints of Fiscal Space	9
3.1 Interest–Growth Differential, Convenience Yield, and the Role of Central Banks	9
3.2 Private Saving Patterns and Fiscal Space	10
3.3 Empirical Analysis	12
3.3.1 Background and empirical motivation.	12
3.3.2 Preliminary work on the series	13
3.3.3 Data	14
3.3.4 Results	15
4 Conclusion	19
Bibliography	20

Introduction

The sustained increase in public debt across advanced economies over recent decades has renewed concerns about the sustainability of sovereign finances and the existence of debt thresholds beyond which fiscal positions become untenable. In OECD countries, average gross public debt rose from around 60 percent of GDP in the early 1990s to well above 120 percent following the COVID-19 pandemic. While high debt levels have historically been associated with concerns about solvency and market access, recent experience suggests that debt dynamics and market outcomes do not always align.

The question of debt sustainability is not new. Early contributions by Ricardo and Domar established the foundations of long-run public debt analysis by emphasizing the interaction between growth, interest rates, and fiscal balances. Modern approaches have formalized these insights through the intertemporal government budget constraint and its empirical counterparts. More recently, fiscal reaction functions à la Bohn and institutional frameworks such as the IMF's Debt Sustainability Analysis have become standard tools for assessing sustainability. Despite their widespread use, these approaches rely on restrictive assumptions regarding fiscal behavior, interest–growth differentials, and market responses, and often yield conflicting assessments across countries with similar debt trajectories. In particular, advanced economies display striking heterogeneity in their ability to sustain high and rising levels of public debt. Countries such as Japan have maintained market access despite debt ratios far exceeding traditional benchmarks, while others have experienced episodes of market stress at much lower levels of indebtedness. These observations have motivated a growing literature emphasizing the concept of fiscal space, defined as the government's capacity to absorb shocks or sustain deficits without triggering adverse market reactions. Fiscal space provides a more flexible and forward-looking perspective on sustainability, incorporating not only debt dynamics but also institutional credibility, investor confidence, and macro-financial conditions.

This paper contributes to the literature in several ways. First, it provides a critical review of the main theoretical and empirical frameworks used to assess sovereign debt sustainability in advanced economies, examining their underlying assumptions and exploring possible explanations for their occasional inconsistencies with empirical evidence and financial markets pricing. Second, it complements this discussion with an analysis of the determinants of fiscal space, highlighting its relevance for understanding discrepancies between model-based assessments and observed outcomes. Finally, the paper presents an empirical analysis focusing on the role of private wealth dynamics in shaping long-term government yields. Using U.S. data, it examines whether increases in net private wealth are associated with lower long-term government bond yields, and whether this relationship depends on the degree of foreign ownership of public debt.

1 Critical Literature Review of the Main Methods to Assess Sustainability

This section reviews the main methodologies used to assess sovereign debt sustainability in advanced economies.

1.1 The Traditional IGBC Framework and Its Limitations

One of the main frameworks in the literature—hereafter referred to as the *traditional approach*—states that public debt is sustainable if the current debt-to-GDP ratio equals the discounted sum of future primary surpluses relative to GDP. This condition is derived from the **Intertemporal Government Budget Constraint (IGBC)**, itself obtained from the government flow budget constraint:

$$d_t = \sum_{j=1}^{\infty} f_{t,t+j} pb_{t+j}, \quad (1)$$

where d_t denotes the debt-to-GDP ratio and pb_{t+j} future primary balances relative to GDP. The discount factor $f_{t,t+j}$ is defined as the **growth-adjusted discount factor**:

$$f_{t,t+n} = \prod_{b=t}^{t+n-1} \left(\frac{1 + g_b}{1 + r_b} \right), \quad (2)$$

with g_b denoting nominal GDP growth and r_b the nominal interest rate on government debt.

Despite its theoretical appeal, the IGBC-based approach faces important conceptual and empirical limitations. First, persistent primary deficits are not exceptional if we look at historical data. For instance, between 1916 and 1995, the United States recorded an average primary deficit of -1.2% of GDP (Bohn, 1998). Empirically, Hakkio and Rush (1991) show that over the period 1950–1988, although government revenues and expenditures were cointegrated, revenues increased by only 68 cents for each additional dollar of spending. This pattern suggests systematic deficit financing through debt issuance and an apparent violation of the IGBC. Yet, this period was not associated with episodes of market distress, raising doubts about the practical relevance of the constraint.

Another limitation is operational. Verifying the IGBC requires knowledge of the entire future paths of primary surpluses, interest rates, and growth—variables that are fundamentally unobservable.

Beyond this feasibility issue, the framework also relies on strong assumptions. In particular, interpreting the IGBC requires the transversality condition—a restrictive version of the Non-Ponzi Game condition—to hold:

$$\lim_{n \rightarrow \infty} f_{t,t+n} d_{t+n} = 0.$$

This assumption appears implausible in advanced economies. Brender, Pisani, and Gagna (2021) emphasize that households display persistent saving behavior over the life cycle, accumulating wealth during working years and decumulating it only gradually during retirement. These savings are predominantly invested in debt instruments. Given limited corporate borrowing capacity, governments effectively act as borrowers of last resort, supplying the safe assets needed to absorb private

savings. In such an environment, there is no economic reason for public debt to converge to zero in the long run, even when it remains sustainable.

The construction of the IGBC also assumes that $r > g$. However, the empirical relevance of this assumption is questionable. Bohn (1995) and Blanchard (2019) document that government borrowing costs have frequently been lower than GDP growth in advanced economies. As shown in Figure 1, in the United States, over the period 1930–2024, nominal GDP growth exceeded the 10-year Treasury yield on average, with $r - g \approx -1.54\%$.

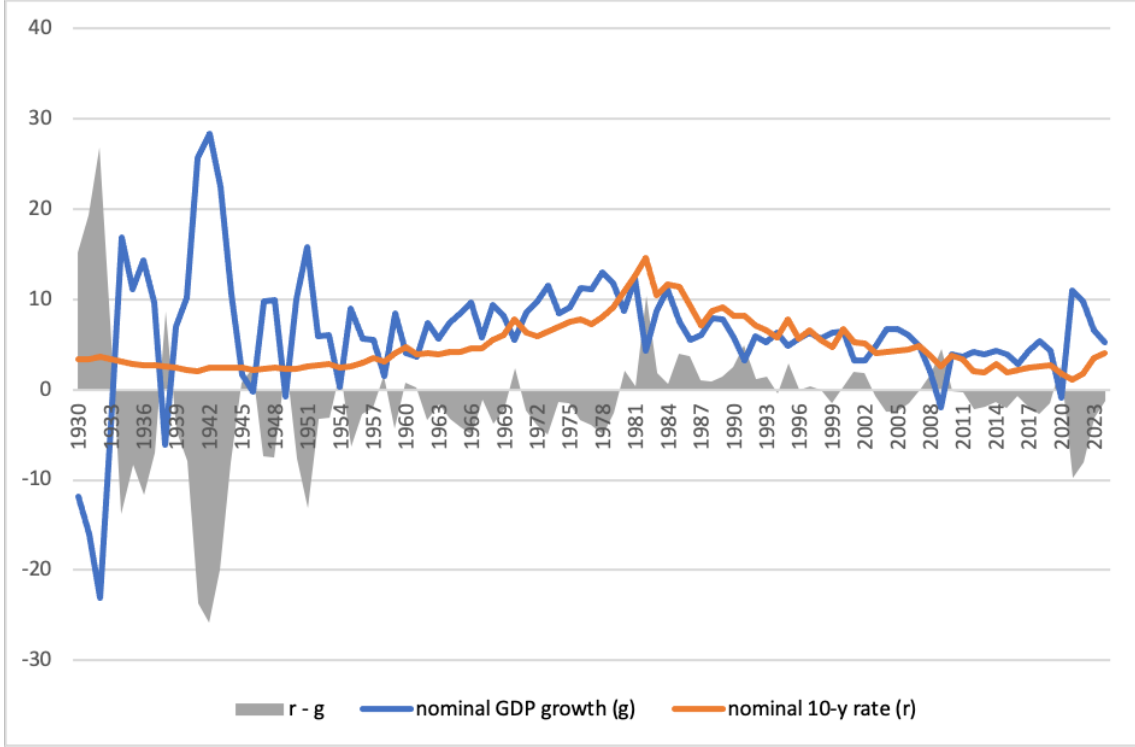


Figure 1: U.S. Nominal GDP Growth Rate and 10-year Bond Rate, 1930–2024

When $r < g$, the growth-adjusted discount factor diverges, invalidating the transversality condition and undermining the construction of the IGBC itself.

1.2 Debt Sustainability under a more realistic framework: Bohn (1995, 1998)

Bohn (1995) rethinks the IGBC by explicitly introducing uncertainty and risk aversion, and by replacing the government borrowing rate-based discount factor with a stochastic discount factor derived from agents' marginal rate of substitution. Assuming a representative agent with time-separable preferences, a constant subjective discount factor $\beta \in (0, 1)$, and complete markets, the appropriate growth-adjusted stochastic discount factor can be written as

$$\tilde{f}_{t,t+j} = E_t \left[u_{t,t+j} \prod_{b=t}^{t+j-1} (1 + g_b) \right], \quad (3)$$

where

$$u_{t,t+j} = \beta^j \frac{U'(PB_{t+j})}{U'(PB_t)}. \quad (4)$$

Here, $U(\cdot)$ denotes the concave utility function of the representative consumer, capturing risk aversion, and $U'(\cdot)$ its marginal utility. PB_{t+j} represents the gross government primary balance at $t+j$.

Under this stochastic discounting, the transversality condition becomes :

$$\lim_{n \rightarrow \infty} E_t[\tilde{f}_{t,t+n} d_{t+n}] = 0$$

, and the corresponding intertemporal government budget constraint can be expressed as

$$d_t = E_t \left[\sum_{j=1}^n \left(\tilde{f}_{t,t+j} pb_{t+j} + \text{cov}(\tilde{f}_{t,t+j}, pb_{t+j}) \right) \right], \quad (5)$$

where d_t denotes the public debt-to-GDP ratio at time t and pb_{t+j} the primary balance (surplus if positive, deficit if negative) relative to GDP at time $t+j$.

Equation (5) highlights the central role of the covariance term between the stochastic discount factor and future primary balances. The intertemporal constraint can be relaxed even when expected primary surpluses are negative, provided the covariance term sufficiently offsets weak expected fiscal fundamentals. Intuitively, if public debt provides insurance-like services—by delivering relatively higher payoffs (here high pb) in states where marginal utility is high—the covariance term is positive, easing the government’s intertemporal constraint. However, in practice, governments typically pursue countercyclical fiscal policies, running deficits in downturns, where discount factor is high. As a result, primary balances tend to be negatively correlated with marginal utility, implying a covariance term that tightens, rather than relaxes, the IGBC.

Bohn’s contribution is therefore not to assert that the covariance term is empirically favorable, but to demonstrate that ignoring it is conceptually incorrect. By explicitly incorporating risk and state contingency, his framework shows that the traditional deterministic IGBC is neither a necessary nor a sufficient condition for fiscal sustainability.

In his 1998 paper, Bohn proposes an operational approach to fiscal sustainability that bypasses the IGBC and instead focuses on observed fiscal behavior. Sustainability is assessed through the systematic response of primary surpluses to changes in the debt-to-GDP ratio.

Fiscal policy is modeled using a fiscal reaction function of the form

$$pb_t = \rho d_t + \mu_t, \quad (6)$$

where pb_t denotes the primary balance relative to GDP, d_t the debt-to-GDP ratio, and μ_t captures other determinants of fiscal policy, including temporary government spending and cyclical conditions. These control factors ensure that the estimated response of primary surpluses to debt is not driven by short-run shocks.

The key parameter is ρ , the fiscal reaction coefficient. When the growth rate exceeds the interest rate ($r < g$), Bohn shows that $\rho > 0$ is a sufficient condition for fiscal sustainability. A positive ρ implies that increases in debt—whether due to recessions, wars, or interest rate shocks—eventually trigger higher primary surpluses, thereby ensuring intertemporal solvency.

A central implication of this framework is that fiscal sustainability does not hinge on the stability of the debt-to-GDP ratio itself. As emphasized by Bohn (1998), a rising debt ratio does not necessarily indicate unsustainable policies, nor does a

stable ratio guarantee sustainability. What matters is whether fiscal authorities actively adjust primary balances in response to debt accumulation.

Accordingly, even in advanced economies where debt ratios have followed upward trends, the empirical observation of a positive fiscal reaction coefficient under $r < g$ can be interpreted as sufficient evidence that fiscal policy remains sustainable. This behavioral criterion stands in contrast to rule-based approaches relying on fixed numerical thresholds, and provides a more flexible and economically grounded assessment of debt sustainability.

However, the fiscal reaction approach remains incomplete, as it abstracts from factors such as market confidence, political economy constraints, and nonlinear fiscal behavior; empirically, this limitation is illustrated by Lamé, Lequien, and Pionnier (2010), who find that Greece satisfied Bohn’s sufficient condition over 1978–2007 while France did not—results starkly contradicted by post-2007 sovereign debt outcomes.

1.3 Stabilization-Based Approaches: Debt Dynamics and IMF DSA

International institutions such as the IMF define public debt sustainability as the government’s ability to stabilize the debt-to-GDP ratio with a high probability over time, taking into account both solvency and liquidity considerations (Blanchard, 2022). This definition relies on the standard debt dynamics equation

$$\Delta d_t \approx (r_t - g_t)d_{t-1} - pb_t, \quad (7)$$

which highlights the central role of the interest–growth differential, often referred to as the snowball effect.

When $r_t > g_t$, debt dynamics are mechanically unfavorable: even with a zero primary balance, a positive initial debt implies an ever-increasing debt-to-GDP ratio. Conversely, a persistently negative $r - g$ can exert a powerful stabilizing force. A simple counterfactual simulation for the United States illustrates this mechanism. Despite running an average primary deficit of about 1.2% of GDP between 1916 and 1995, U.S. debt increased only from roughly 2.7% to 54% of GDP. Holding the same primary deficit but assuming $r - g = 0$ would have led to a counterfactual debt ratio close to 99% in 1995. Hence, the historically negative $r - g$ appears to have averted around 45 percentage points of additional debt accumulation (own calculations).

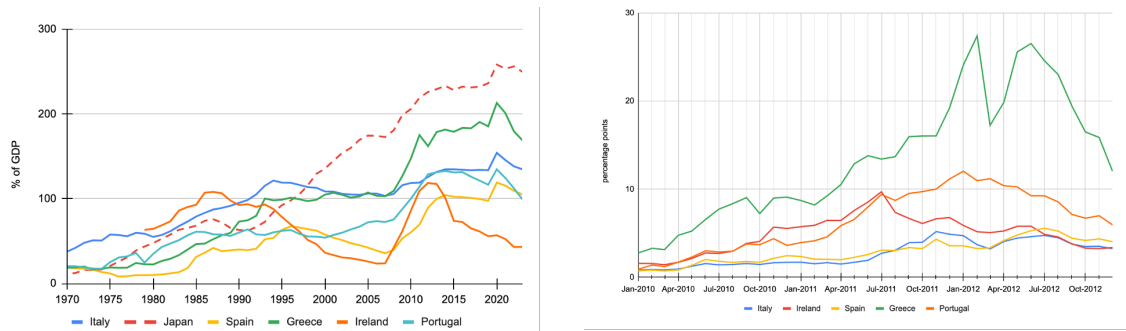
The IMF operationalizes this stabilization-based view through the Debt Sustainability Analysis (DSA). Rather than relying on long-run solvency conditions such as the IGBC, the DSA projects debt dynamics forward under baseline and adverse scenarios and evaluates whether debt is likely to stabilize given assumed paths for growth, interest rates, and primary balances. Like Bohn’s fiscal reaction approach, the DSA focuses on fiscal behavior rather than debt levels per se: a rising debt ratio may still be considered sustainable if fiscal policy is expected to adjust sufficiently to ensure future stabilization.

Despite its practical relevance, the DSA faces two persistent challenges. First, fiscal adjustments interact endogenously with growth: consolidation may improve the primary balance while simultaneously depressing economic activity, potentially worsening debt dynamics, as illustrated by the Greek experience after 2010. Second, sustainability assessments are highly sensitive to baseline assumptions, particularly long-run growth projections. Small differences in assumed growth rates can lead to radically different conclusions, as shown by the divergent IMF and European Commission DSAs for Greece in 2015.

2 Fiscal Space and Country-Specific Debt Limits

2.1 Fiscal Space: Motivation and Core Concept

The application of standard sustainability frameworks to real-world cases reveals striking divergences between debt dynamics and market outcomes. Comparable upward debt trajectories have often resulted in radically different assessments of sustainability and investor reactions. A particularly revealing contrast, illustrated by Figure 2 is provided by Japan and several euro area countries during the sovereign debt crisis. Japan experienced a sustained rise in its debt-to-GDP ratio, from about 50% in 1990 to nearly 260% by 2020, without facing market stress. By contrast, during 2010–2012, several euro area countries experienced sharp spikes in sovereign spreads amid fears of fiscal unsustainability, despite significantly lower debt levels.



(a) Gross Public Debt, Percent of GDP, 1970–2023

(b) 10-year Government Bond Spreads over German Bunds, 2010–2012

Figure 2: Debt-to-GDP Ratios and Yields in Europe and Japan

This contrast highlights the limitations of interpreting sustainability through uniform debt thresholds, such as the Maastricht Treaty’s 60% of GDP benchmark. Several countries, including France, exceeded this limit for prolonged periods without market pressure, while Japan reached levels far beyond what had traditionally been considered compatible with sustainability and yet retained stable market access.¹ These observations suggest that sovereign risk is not determined by debt levels alone, but by country-specific factors shaping investors’ expectations and tolerance for public debt.

This insight motivates the concept of fiscal space, defined as the room a government has to increase spending or reduce taxes without jeopardizing debt sustainability or triggering a loss of market confidence. More formally, fiscal space represents the distance between the current debt-to-GDP ratio and a country-specific debt limit beyond which rollover risk, liquidity stress, or self-fulfilling debt crises may emerge. Fiscal space therefore captures both solvency and liquidity dimensions of sustainability and provides an explicit measure of a government’s capacity to absorb adverse shocks or delay fiscal adjustment.

The relevance of fiscal space becomes particularly clear in environments characterized by uncertainty and nonlinear fiscal behavior. While Bohn’s fiscal reaction framework assumes a stable and continuous adjustment of primary balances to rising debt, this relationship may weaken beyond a critical point. Political constraints,

¹The idea of a universal debt threshold drew support from Reinhart and Rogoff (2010), who found that growth tends to slow when public debt exceeds 90% of GDP. Their findings, however, were later challenged for data omissions and methodological flaws (Herndon, Ash, and Pollin, 2013), and the 90% figure, often interpreted as a debt limit, was not presented as such by the authors.

social fatigue, or macroeconomic rigidities may prevent governments from sustaining fiscal consolidation at very high debt levels, a phenomenon often referred to as fiscal fatigue. As a result, a country may appear to have behaved sustainably in the past, yet become vulnerable once its fiscal space is exhausted.

2.2 Fiscal Fatigue, Debt Limits, and Market Expectations

To formalize these ideas, Ostry, Ghosh, Kim, and Qureshi (2010) develop a stylized model of fiscal space, illustrated on Figure 3, that links debt sustainability to the interaction between fiscal behavior and debt servicing costs. In this framework, the primary balance increases with the debt-to-GDP ratio, reflecting a responsible fiscal response consistent with a positive fiscal reaction coefficient. However, at higher debt levels, this response weakens as fiscal adjustment becomes increasingly difficult. At the same time, the interest–growth differential generates a snowball effect that raises debt servicing costs proportionally with the level of debt if $r - g > 0$.

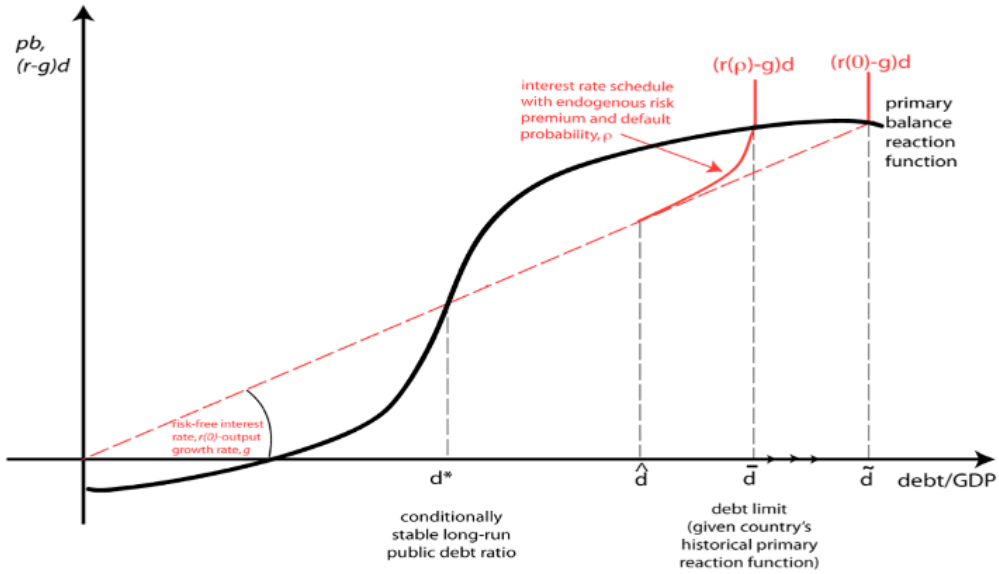


Figure 3: Determination of Debt Limit

The interaction of these two forces gives rise to a critical debt threshold ($\tilde{d}$). Below this threshold, the economy converges toward a stable long-run debt ratio (d^*). Beyond $\tilde{d}$, the primary surplus becomes insufficient to offset the snowball effect, and debt dynamics turn explosive. This threshold defines the debt limit. The model is extended to incorporate uncertainty and investor risk aversion. In the stochastic version, adverse shocks to growth, interest rates, or fiscal outcomes increase perceived default risk. Anticipating this risk, investors demand a higher risk premium as debt rises, causing the effective interest rate to increase endogenously with the debt level. As a result, the snowball effect steepens and becomes more convex as debt approaches its perceived limit. This mechanism compresses fiscal space before the structural debt limit ($\tilde{d}$) is reached and introduces a forward-looking dimension to sustainability, shaped by market expectations and confidence. Formally: $FiscalSpace = \tilde{d} - d_t$. When fiscal space is ample, governments can temporarily deviate from fiscal discipline without triggering market stress. Once it is depleted, even small adverse shocks may provoke a sudden loss of confidence.

Empirical extensions of this framework further highlight the country-specific nature of fiscal space. Ghosh et al. (2013), extending the model to account for fiscal fatigue using data from 1985–2007, find that for several countries—including Greece, Italy, Portugal, Iceland, and Japan—no finite debt limit can be statistically identified, suggesting that fiscal space had effectively been exhausted. With the notable exception of Japan, all of these countries experienced severe debt crises between 2008 and 2012. The same study also documents substantial heterogeneity in estimated debt limits across countries, with values around 230% of GDP for Norway, 170% for Germany and the United States, and approximately 160% for France. Given projected debt ratios well below these limits at the time, the implied fiscal space varied widely across countries, underscoring that debt sustainability is inherently context-dependent.

An important implication of the fiscal space framework is that debt sustainability can deteriorate well before debt dynamics become formally unsustainable. Under uncertainty, fiscal space is vulnerable to a range of fragilities that can abruptly alter market perceptions and compress the margin for adjustment.

A first source of fragility arises from contingent liabilities, defined as obligations that are not reflected in the current fiscal balance but may materialize depending on future events. These liabilities can accumulate gradually, as in the case of demographic pressures that raise pension and healthcare spending, or climate-related risks that increase the frequency and fiscal cost of natural disasters. They can also emerge suddenly, for instance through government guarantees, financial sector bailouts, or emergency interventions following large shocks. When such liabilities materialize, they can rapidly deplete fiscal space by pushing debt closer to its perceived limit.

A second fragility stems from nonlinear market repricing. Investor perceptions of sovereign risk do not adjust smoothly with fundamentals: borrowing costs may remain low over a wide range of debt levels, but rise sharply once confidence erodes. This nonlinearity implies that fiscal space can disappear suddenly, even if the observed debt trajectory appears stable. Expectations about future growth, fiscal commitment, or institutional credibility play a central role in triggering such repricing episodes.

These mechanisms are illustrated by the United Kingdom’s experience in 2022. In September 2022, the announcement of a large, unfunded fiscal package—combined with the absence of an independent fiscal assessment—led markets to reassess the government’s fiscal credibility. Despite no immediate change in the debt-to-GDP ratio, yields on long-term government bonds rose sharply and the currency depreciated, forcing the central bank to intervene to stabilize financial markets. In the language of the fiscal space framework, the UK episode reflects a sudden leftward shift in the perceived debt limit, driven by higher risk premia and downward revisions in expected future primary balances. Fiscal space narrowed abruptly, not because of higher debt, but because of a loss of confidence.

Fiscal space is shaped by expectations and credibility rather than a static debt threshold. Similar debt levels can therefore lead to very different sustainability outcomes. This shifts attention from short-run indicators toward the structural determinants of fiscal space that matter for debt sustainability in advanced economies.

3 Long-Term Determinants and Constraints of Fiscal Space

3.1 Interest–Growth Differential, Convenience Yield, and the Role of Central Banks

A central long-term determinant of fiscal space in advanced economies is the persistence of an interest–growth differential such that $r < g$. As emphasized by Blanchard (2019), when the government borrowing rate remains durably below the growth rate, the fiscal effort required to stabilize debt is mechanically reduced, thereby expanding fiscal space. However, the empirical prevalence of $r < g$ raises a deeper question: why are governments able to borrow at such low rates for extended periods?

A growing literature argues that government bonds are valued not only for their promised cash flows, but also for the liquidity and safety services they provide to investors. Building on Woodford (1990) and Holmström and Tirole (1998), recent contributions emphasize that public debt delivers a *convenience yield*: investors are willing to accept lower returns in exchange for holding assets that are liquid, safe, and reliable stores of value in bad states of the world (Berentsen and Waller, 2018; Brunnermeier, Merkel, and Sannikov, 2021; Reis, 2021). This interpretation is particularly relevant for advanced economies whose sovereign bonds serve as global or regional safe assets.

Accounting for this service flow fundamentally alters the traditional intertemporal interpretation of public debt. In the standard IGBC framework, the market value of government debt must be backed by the expected present value of future primary surpluses:

$$\text{Value of debt stock} = E[PV(\text{future primary surpluses})]. \quad (8)$$

Once liquidity and safety services are taken into account, however, the valuation of public debt includes an additional component:

$$\text{Value of debt stock} = E[PV(\text{future primary surpluses})] + E[PV(\text{future service flow})]. \quad (9)$$

As emphasized by Willems and Zettelmeyer (2022), this extension implies that intertemporal solvency no longer requires the transversality condition to hold in its classical form. Instead, solvency can be maintained as long as the terminal value of debt is matched by the discounted value of the service flow it provides. In this sense, public debt may remain sustainable even in the presence of persistent primary deficits, provided that the liquidity and safety services embedded in government bonds retain sufficient value.

This perspective offers a reinterpretation of earlier stochastic approaches to debt sustainability. In particular, Bohn’s (1995) covariance term between future primary balances and the stochastic discount factor can be understood as an early attempt to capture the state-contingent value of public debt as a safe and liquid asset. When government bonds deliver higher value in adverse states of the world, their service flow effectively relaxes the intertemporal constraint faced by the government.

Crucially, the service flow component is neither unlimited nor guaranteed. Its magnitude depends on institutional credibility, market confidence, and the perceived safety of public debt. Jiang et al. (2019) estimate that, for the United States, the service flow associated with Treasuries amounted to roughly 65% of GDP on average, highlighting its quantitative relevance. However, if confidence in a government’s fiscal or monetary institutions erodes, the convenience yield can shrink or disappear, sharply reducing fiscal space.

Central bank credibility plays a key role in sustaining this mechanism. A credible central bank enhances the safety of government debt by maintaining control over inflation expectations and supporting countercyclical monetary policy. It also enhances liquidity by fostering deep and resilient sovereign bond markets and, when necessary, acting as a backstop during periods of stress. Willems and Zettelmeyer (2022) conceptualize this credibility as a fiscal asset: accumulated over time, it can be drawn upon to support public debt sustainability, but it is exhaustible if overused.

Taken together, these elements suggest that long-term fiscal space in advanced economies is shaped not only by fiscal fundamentals, but also by the structural demand for safe and liquid assets, the credibility of monetary institutions, and the persistence of excess savings. These factors help explain why some countries, such as Japan, can sustain high and rising debt levels at relatively low yields, while others face binding constraints at much lower levels of public debt, such as some eurozone countries during the Sovereign Debt crisis.

3.2 Private Saving Patterns and Fiscal Space

Structural private saving behavior plays a key role in shaping fiscal space in advanced economies. As emphasized by Brender, Pisani, and Gagna (2021), households exhibit persistent life-cycle saving patterns, accumulating wealth over long horizons and allocating a large share of these savings to low-risk financial assets, notably government bonds. This behavior generates a structural excess of desired saving relative to private investment opportunities.

When these excess savings cannot be absorbed by the private sector, aggregate demand remains persistently weak, giving rise to disinflationary or deflationary pressures. In such environments, monetary policy may become constrained, particularly when interest rates approach the zero lower bound and transmission mechanisms weaken. Lower policy rates may fail to stimulate investment or consumption sufficiently, leaving excess savings largely unabsorbed.

In this context, the government effectively acts as a borrower of last resort. By issuing public debt, it supplies the safe and liquid assets demanded by households while converting excess private savings into effective demand through public spending or transfers. As argued by Brender, Pisani, and Gagna (2021), this role is essential to stabilize output and prevent deflationary spirals when private demand is structurally insufficient. High and persistent public debt can therefore emerge as an equilibrium outcome rather than a symptom of fiscal irresponsibility.

This mechanism has direct implications for fiscal space. First, this accumulated private wealth through savings sustain strong demand for government bonds, allowing sovereigns to borrow at low interest rates even at high debt levels. Second, the deflationary pressure generated by excess savings enables central banks to maintain accommodative monetary policy without undermining inflation credibility, further supporting low borrowing costs. Together, these forces expand fiscal space by relaxing financing constraints on the government.

However, this configuration is not permanent. If private saving behavior weakens—for instance due to demographic changes or declining precautionary motives—the government must adjust its borrowing accordingly. Maintaining high public debt in the absence of excess savings may generate inflationary pressure, trigger monetary tightening, and ultimately compress fiscal space. This reinforces Bohn’s (1998) insight that neither rising nor stable debt-to-GDP ratios are sufficient indicators of

sustainability in isolation.

The Japanese experience illustrates this logic. Since the 1990s, persistent excess savings (4), accumulated private wealth, and weak private demand have justified sustained government borrowing without triggering inflation or market stress. Periods in which part of these savings were absorbed externally, notably through improved net exports, reduced the need for further debt accumulation. Excess private savings are observed for Germany and France, especially since the 2000s.

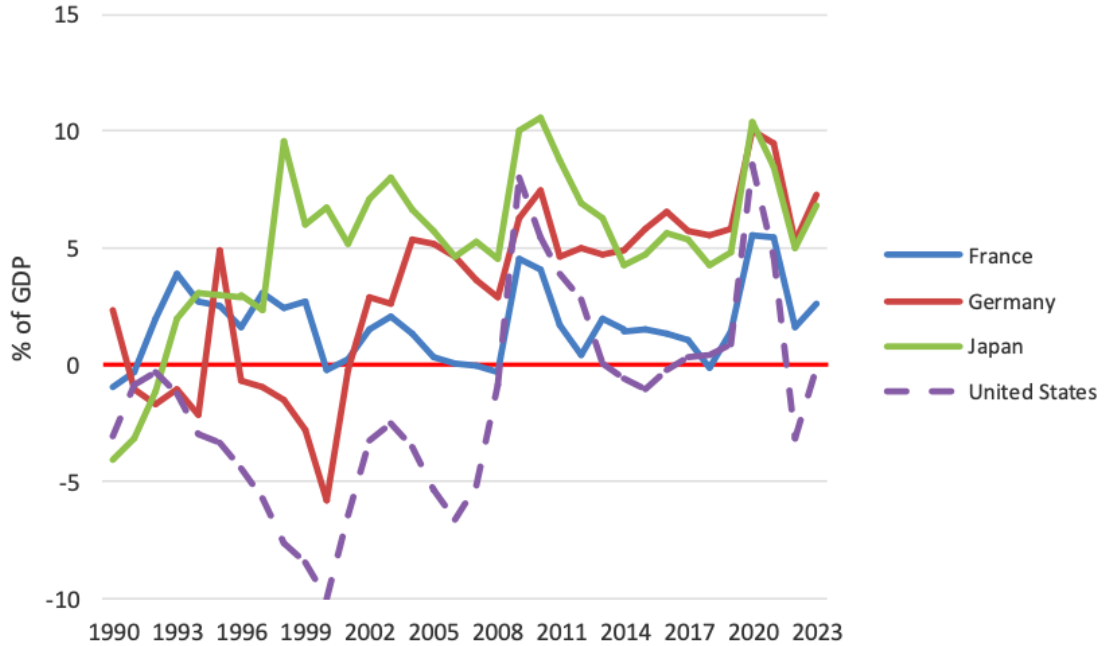


Figure 4: Private Saving - Domestic Investment, 1990-2023

Overall, fiscal space appears tied to private saving and wealth dynamics. In this spirit, Tokuoka (2010), provides empirical evidence that Japan's high domestic absorption of government debt contributed to lowering long-term interest rates and expanding fiscal space. He emphasizes that the sustained increase in household wealth, combined with a strong preference for domestic government bonds, exerted downward pressure on government borrowing costs.

To assess this mechanism, Tokuoka estimates a regression of the 10-year Japanese government bond yield on the debt-to-GDP ratio, net private financial wealth relative to GDP (used as a proxy for domestic debt absorption capacity) and a set of macroeconomic control variables. His results show a negative and statistically significant coefficient on net private financial wealth-to-GDP, consistent with the interpretation that higher financial wealth and saving capacity reduce sovereign yields by increasing demand for government debt.

This empirical approach motivates the econometric analysis developed in the next section. Building on Tokuoka's framework, we examine whether similar wealth-driven mechanisms operate in the United States, while adapting the specification to account for differences in financial structure and investor base.

3.3 Empirical Analysis

3.3.1 Background and empirical motivation.

Our central hypothesis is that increases in net private wealth—particularly among domestic households—support demand for safe assets and help contain long-term government bond yields, even as the debt-to-GDP ratio rises in the United States (Figure 5).

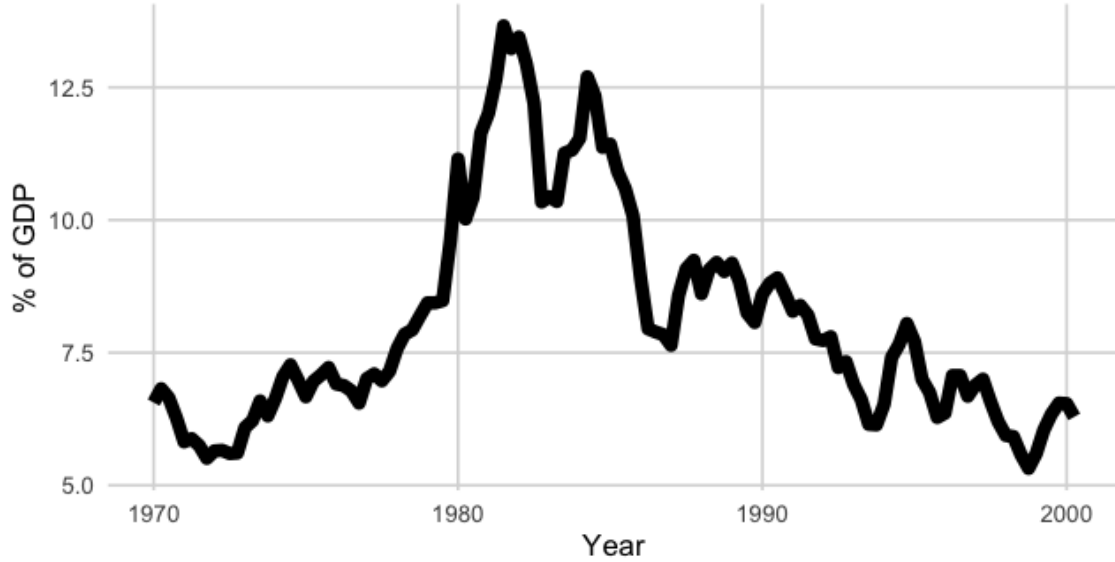


Figure 5: Federal Debt Relative to GDP, 1970-2025

Importantly, changes in the share of U.S. public debt held by foreign investors are likely to affect the strength of this relationship. Until the late 1990s, the share of U.S. government debt held by foreign investors remains relatively limited, while net private wealth relative to GDP increases gradually. After 2000, foreign holdings of U.S. public debt rise sharply, coinciding with a marked acceleration in net private wealth accumulation. This structural change suggests that the relationship between private wealth, debt ownership, and long-term interest rates may differ across periods.

Given data availability constraints prior to 1970 and the evident shift in debt ownership around the early 2000s, the empirical analysis is conducted using a split-sample approach, comparing the periods 1970–2000 and 2000–2023. This design allows us to assess whether the impact of private wealth on long-term yields—and, by extension, on fiscal space—is affected by the degree of domestic versus foreign ownership of government debt.

Figure 6 provides descriptive evidence motivating the empirical strategy.

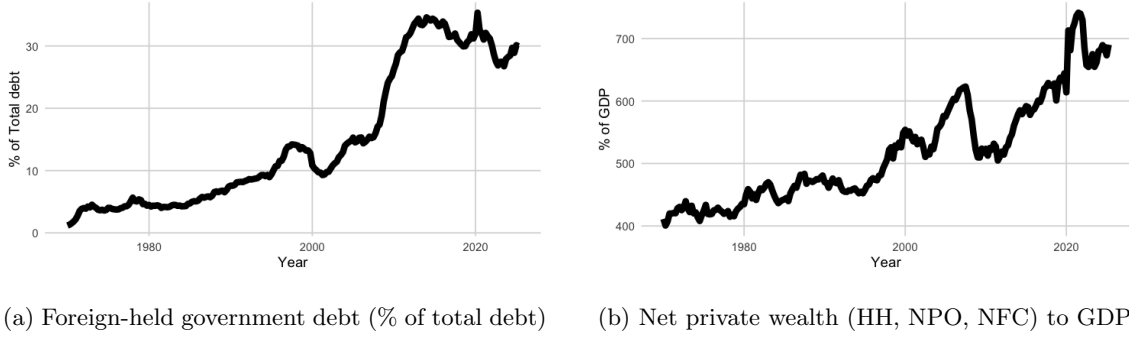


Figure 6: Foreign debt share and net private wealth in the United States (1970–2023).

3.3.2 Preliminary work on the series

The empirical analysis builds on the framework proposed by Tokuoka (2010), in which long-term government bond yields are regressed on net private financial wealth, controlling for a set of macroeconomic fundamentals. Accordingly, the baseline specification relates long-term yields to net private wealth relative to GDP, augmented with a vector of control variables: core sticky inflation, real GDP growth, the short-term interest rate, the share of government debt held by foreign investors, and the public debt-to-GDP ratio.² Formally, the empirical specification can be written as:

$$\text{Yield}_t = \alpha + \beta_1 \text{DebtGDP}_t + \beta_2 \text{NetWealthGDP}_t + \Gamma' X_t + \varepsilon_t, \quad (10)$$

where yield_t denotes the long-term government bond yield, DebtGDP_t is the public debt-to-GDP ratio, NetWealthGDP_t denotes net private wealth relative to GDP, and X_t is a vector of control variables including inflation, real GDP growth, the short-term interest rate, foreign debt share of the government debt, and public debt relative to GDP.

The first three controls follow Tokuoka (2010) and capture inflationary pressures, business cycle conditions, and the monetary policy stance. The inclusion of public debt relative to GDP is relevant given the level of debt-to-GDP could influence the pressure on the yield. The inclusion of foreign debt holdings reflects specific features of the U.S. economy, notably its safe-haven status and the prominent role of international investors in Treasury markets.

A preliminary inspection of the data suggests that several series exhibit strong trending behavior. In particular, net private wealth relative to GDP, the public debt-to-GDP ratio, and the share of foreign-held government debt display persistent upward movements over the sample period. Formal unit root tests confirm this visual evidence. Augmented Dickey–Fuller (ADF) tests fail to reject the null hypothesis of a unit root for these variables when tested in levels, regardless of whether a constant or a deterministic trend is included. By contrast, first differences of these series are found to be stationary, indicating that they are integrated of order one.

Table 1 reports the ADF test results for the non-stationary variables. ADF statistics for the remaining variables are available upon request.³

²For the United States, net private wealth is measured using total net worth rather than net financial wealth. This choice is primarily driven by data availability, as long and consistent time series of net financial wealth could not be identified for the United States. Although conceptually different from net financial wealth, total net worth remains economically meaningful, as it reflects the private sector’s ability to absorb public debt through its aggregate balance sheet.

³Short-term interest rates are found to be stationary in levels, while long-term yields become stationary after first differencing.

Table 1: ADF unit root tests summary

Variable	Level: p-value	Level conclusion	$\Delta 1$: p-value	Integration order
Public debt to GDP	0.639	Non-stationary	< 0.01	I(1)
Foreign debt share	0.338	Non-stationary	0.031	I(1)
Net wealth to GDP	n.s.	Non-stationary	< 0.01	I(1)

Tokuoka (2010) proceeds by estimating the model in levels and addressing non-stationarity through an Engle–Granger cointegration approach. While such a strategy is feasible, it relies on the existence of a stable long-run equilibrium relationship between yields and private wealth. In the present context, this assumption is not innocuous. Given the limited sample size, the presence of structural breaks, and the strong trending behavior of key variables, estimating the model in levels exposes the analysis to the risk of spurious regression.

Instead, this paper adopts a more conservative approach by transforming the variables into log growth rates, to address non-stationarity concerns directly. While this transformation precludes the identification of long-run equilibrium relationships, it enables a reliable analysis of short- to medium-term dynamics between changes in long-term yields and changes in private wealth.

Accordingly, the specification can be expressed as following:

$$\Delta \log(\text{yield}_t) = \alpha + \beta \Delta \log(\text{NetWealthGDP}_t) + \Gamma' K_t + \varepsilon_t, \quad (11)$$

where K_t denotes the vector of stationarized control variables, which are transformed using the same log-growth approach.

3.3.3 Data

All macroeconomic and financial variables are obtained from the FRED database.⁴ The analysis uses quarterly data over the period 1970–2023, with a split sample between 1970–2000 and 2000–2023, reflecting both data availability and a structural change in the ownership of U.S. public debt.

Long-term government bond yields. For the period 1970–2000, long-term government bond yields are measured using the series LTGOVTBD, which corresponds to the unweighted average yield on all outstanding U.S. Treasury bonds with a remaining maturity of at least ten years and that are neither due nor callable in the near term. From 2000 onward, long-term yields are measured using the constant-maturity 10-year Treasury yield (DGS10), which has become the standard benchmark for long-term interest rates. The two series are spliced to construct a consistent measure of long-term government borrowing costs.

Debt-to-GDP. Public debt is measured using the federal debt-to-GDP ratio (GFDEGDQ188S), which captures the outstanding stock of U.S. federal government debt relative to nominal GDP.

Net private wealth. Net private wealth is constructed as the sum of household and nonprofit organizations’ net worth (TNWBSHNO) and nonfinancial corporate net

⁴Operated by the Federal Reserve Bank of St. Louis

worth (TNWMVBSNNCB). Net private wealth is scaled by nominal GDP to obtain net private wealth relative to GDP.

Foreign ownership of government debt. The share of U.S. government debt held by foreign investors is measured using the series HBFIGDQ188S.

Short-term interest rate. The monetary policy stance is proxied by the 3-month Treasury bill rate (IR3TCD01USM156N), measured on a discount basis. This variable controls for short-term interest rate movements.

Macroeconomic controls. Nominal GDP (GDP) is used both as a scaling variable and to compute growth rates. Real economic activity is captured by real GDP(GDPC1) growth.

Inflation dynamics are proxied by a measure of core sticky inflation (CORE-STICKM159SFRBATL).

Instruments. Personal dividend income (PDI), housing starts (HOUST), the employment-to-population ratio (EMRATIO), and Real Residential Property Prices (QUSR628BIS) are used as instruments for net private wealth in the instrumental variable specification.

All variables are converted to quarterly frequency.

3.3.4 Results

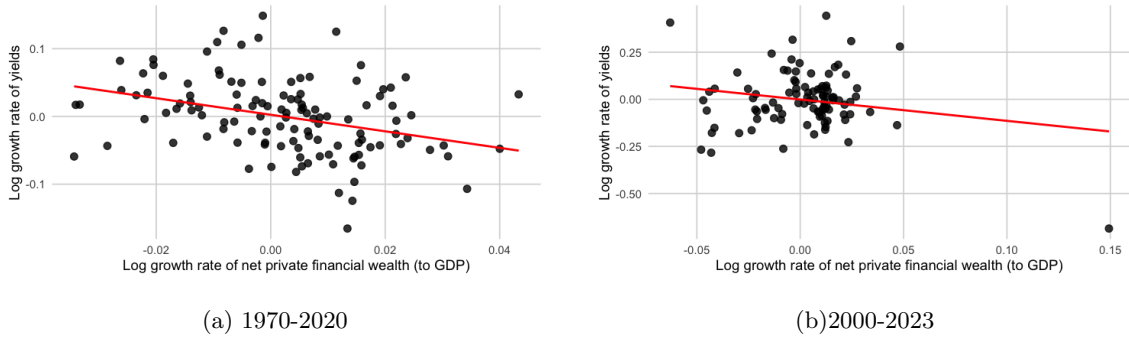


Figure 7: Log-Growth Relationship Between Net Private Wealth and Long-Term Government Yields in the United States (1970–2023).

Figure 7 suggests a negative relationship between the logarithmic growth rate of net private wealth and the one of long-term government bond yields in the United States. This relationship appears visually stronger over the period 1970–2000 than over 2000–2023, consistent with the initial intuition that the wealth channel operates more forcefully when government debt is predominantly absorbed domestically.

This visual pattern is confirmed by the baseline regression results reported in Table 2.

1970-2000

OLS In specification (1), covering the period 1970–2000 with a restricted set of controls, the estimated coefficient on the growth rate of net private wealth is negative

Table 2: Net private wealth, debt ownership and long-term yields

	1970–2000			2000–2023	
	(1)	(2)	(3)	(4)	(5)
	OLS	OLS	IV	OLS	OLS
$\Delta \log(\text{Net wealth}/GDP)$	−0.697*** (0.233)	−0.620** (0.247)	−1.560* (0.938)		0.245 (0.589)
$\Delta \log(\text{Debt}/GDP)$		−0.251 (0.222)			−0.607 (1.061)
$\Delta \log(\text{Foreign debt share})$	−0.077 (0.048)	−0.066 (0.045)	−0.061 (0.071)	−0.878*** (0.322)	−0.735** (0.360)
Core sticky inflation	0.00399*** (0.00133)	0.00408*** (0.00147)	0.00420*** (0.00141)	0.00996 (0.01248)	0.01029 (0.01223)
$\Delta \log(\text{Short-term rate})$	0.233*** (0.0367)	0.225*** (0.0364)	0.206*** (0.0440)	0.0915 (0.0746)	0.0911 (0.0795)
Real GDP growth		0.00039 (0.00405)		0.0242** (0.0119)	0.0167 (0.0196)
Observations	121	121	121	96	96
R^2	0.440	0.445	0.386	0.277	0.281
ADF residuals	I(0)	I(0)	—	I(0)	I(0)
Weak IV (p-value)	—	—	0.043	—	—
Sargan p-value	—	—	0.474	—	—

Notes: The dependent variable is the growth rate of long-term government bond yields. Columns (1), (2), (4), and (5) report OLS estimates with Newey–West (lag 1) heteroskedasticity and autocorrelation consistent standard errors in parentheses. Column (3) reports instrumental variable estimates with standard errors from `ivreg`. ADF tests on residuals confirm stationarity in OLS specifications. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

and statistically significant. Quantitatively, a 1 percent increase in net private wealth relative to GDP is associated with a 0.697 percent decrease in long-term government bond yields in relative terms, which corresponds to a few basis points depending on the initial level of yields.

This result is economically consistent with the mechanism emphasized by Brender, Pisani, and Gagna (2021), whereby an expansion in private balance sheets increases demand for safe public debt, thereby compressing yields.

The estimated coefficient remains highly robust when standard errors are corrected using the Newey–West procedure, indicating that the result is not driven by serial correlation or heteroskedasticity. Moreover, extending the set of control variables in specification (2) does not materially affect the estimated impact of private wealth, which remains close in magnitude (−0.620) and statistically significant.

Instrumental Variable Specification Despite these robust OLS results, potential endogeneity concerns remain. For instance, reverse causality may arise if declining long-term interest rates mechanically increase asset valuations and hence measured private wealth. Also, unobserved macro-financial factors could jointly influence both wealth accumulation and interest rates.

To address these concerns, an instrumental variable approach is implemented in specification (3). Net private wealth is instrumented using four variables: the employment-to-population ratio(EMRAT), housing starts(HOUST), personal dividend income(DIV), and the inflation on real residential prices (RES). Employment-to-population ratio captures labor market conditions and aggregate income dynam-

⁴For small variations, log differences can be interpreted as approximate growth rates.

ics, which directly affect household balance sheets through wage income, savings capacity, and pension contributions, thereby influencing the accumulation of private wealth. Housing starts reflects housing market activity and residential investment, which affect household net worth through housing-related wealth effects and portfolio rebalancing toward financial assets. Dividend income captures capital income flows accruing to households and is directly linked to equity valuations and wealth accumulation. Instead, their influence on yields is expected to operate indirectly through their effect on private balance sheets and portfolio demand for safe government debt. Inflation in real residential prices constitutes the one of the largest component of household balance sheets, so changes in real house prices directly affect wealth accumulation.

At the same time, these variables are unlikely to affect long-term government bond yields directly once standard macroeconomic controls—such as inflation, output growth, and short-term interest rates—are included.

The two-stage least squares specification is given by:

$$\Delta \log(\text{NetWealth}_t) = \pi_0 + \pi_1 \text{EMRAT}_t + \pi_2 \text{HOUST}_t + \pi_3 \text{DIV}_t + \pi_4 \text{RES}_t + \Pi' K_t + u_t, \quad (12)$$

$$\Delta \log(\text{yield}_t) = \alpha + \beta \Delta \log(\widehat{\text{NetWealth}}_t) + \Gamma' K_t + \varepsilon_t, \quad (13)$$

where K_t denotes the vector of control variables stationarized using the same log-growth approach.

Diagnostic tests support the validity of the IV strategy. The first-stage weak instrument p-value indicates that the instruments are highly relevant. Furthermore, the Sargan overidentification test fails to reject the null hypothesis of instrument exogeneity. Importantly, the IV estimates confirm the negative and statistically significant (at 10%) effect of private wealth on long-term yields over the period 1970–2000, and within a more negative magnitude (-1.56). The precision on the magnitude could be improved, but the negative signal is well identified, which is what we were looking for. Thus, even after accounting for potential endogeneity, increases in net private wealth relative to GDP are associated with lower long-term government bond yields in the short to medium run.

Taken together, these results reinforce the interpretation of a domestic wealth-driven demand channel for safe government debt in the US.

2000-2023

OLS Results differ markedly over the 2000–2023 period. In specification (5), estimated by OLS with alternative sets of control variables, the coefficient on net private wealth becomes statistically insignificant, while the share of government debt held by foreign investors emerges as a significant determinant of long-term government bond yields.

Quantitatively, a 1 percent increase in the foreign debt share is associated with an approximately 0.735 (0.878 in OLS (4) with less controls) percent decrease in long-term government bond yields in relative terms. This result is consistent with the macro-financial environment characterizing the post-2000 period, which includes two major global crises (the Global Financial Crisis of 2007–2008 and the COVID-19 shock) and a sharp rise in foreign holdings of U.S. public debt.

Over this period, U.S. Treasuries increasingly served as the global safe asset, reflecting the United States' so-called *exorbitant privilege*. Heightened foreign demand for U.S. government debt likely exerted sustained downward pressure on long-term yields, allowing the U.S. government to finance large deficits at relatively low cost.

In this context, the effect of domestic private wealth on yields appears to be overshadowed by global safe-haven demand, even though domestic wealth continued to expand.

These findings suggest that the relative importance of domestic versus foreign demand channels in government bond markets is time-varying and depends on the international role of sovereign debt.

Endogeneity concerns We acknowledge potential endogeneity concerns affecting the share of foreign-held U.S. government debt, notably due to reverse causality—higher yields may attract foreign capital—as well as omitted global financial factors. As a robustness exercise, we explored an instrumental variable strategy using the USD/JPY exchange rate, which plausibly influences foreign holdings of U.S. Treasuries through valuation effects and portfolio rebalancing, thereby satisfying the relevance condition. The exogeneity of this instrument can be reasonably motivated under the assumption that exchange rate movements primarily reflect shocks to short-term interest rate differentials, as implied by uncovered interest parity, rather than directly affecting long-term U.S. bond yields. However, the resulting IV specification exhibits very weak explanatory power, limiting its economic interpretability. For these reasons, the IV approach is not retained in the baseline analysis, and the OLS results remain the primary reference for assessing the role of foreign debt ownership in shaping U.S. long-term government bond yields. The search for more convincing instruments is left for future work.

4 Conclusion

The theoretical analysis suggests that sovereign debt sustainability in advanced economies is inherently country-specific. Differences in fiscal space, shaped by interest–growth differentials, fiscal behavior, and institutional features, call for caution when applying overly general frameworks such as the intertemporal government budget constraint or simple fiscal reaction functions, which abstract from these country-specific factors. Institutional approaches such as the IMF’s Debt Sustainability Analysis offer a more comprehensive perspective, even if they remain imperfect and sensitive to underlying assumptions.

Complementing this analysis, the empirical results indicate that increases in net private wealth are associated with lower long-term government bond yields when public debt is predominantly domestically held, consistent with a demand-for-safe-assets channel. This relationship weakens as foreign ownership becomes more important. A limitation of this study is its focus on a single country, driven by data availability constraints. Extending the analysis to other advanced economies would help assess the generality of this mechanism and is left for future research, with the United States offering a natural starting point given the breadth of available open-source data.

References

- [1] Berentsen, A. and Waller, C.J. (2018). Liquidity: A new monetarist perspective. *Journal of Economic Literature*, 56(2), pp. 479–500.
- [2] Blanchard, O. (2019). Public debt and low interest rates. *American Economic Review*, 109(4), pp. 1197–1229.
- [3] Bohn, H. (1995). The sustainability of budget deficits in a stochastic economy. *Journal of Money, Credit and Banking*, 27(1), pp. 257–271.
- [4] Bohn, H. (1998). The behavior of U.S. public debt and deficits. *Quarterly Journal of Economics*, 113(3), pp. 949–963.
- [5] Brender, A., Pisani, F. and Gagna, É. (2021). *The Macroeconomics of Debt: Europe’s blind spot*. CEPS.
- [6] Brunnermeier, M.K., Merkel, S. and Sannikov, Y. (2021). Debt as safe asset. *American Economic Review*, 111(5), pp. 1476–1504.
- [7] Ghosh, A.R., Kim, J.I., Mendoza, E.G., Ostry, J.D. and Qureshi, M.S. (2013). Fiscal fatigue, fiscal space and debt sustainability in advanced economies. *Economic Journal*, 123(566), pp. F4–F30.
- [8] Hakkio, C.S. and Rush, M. (1991). Is the Budget Deficit “Too Large?” *Review of Economics and Statistics*, 73(3), pp. 429–445.
- [9] Holmström, B. and Tirole, J. (1998). Private and public supply of liquidity. *Journal of Political Economy*, 106(1), pp. 1–40.
- [10] Jiang, Z., Lustig, H., Van Nieuwerburgh, S. and Xiaolan, M. (2019). The service flow of government debt. *Journal of Monetary Economics*, 104, pp. 1–14.
- [11] Lomé, E. and Léquien, M. (2010). Public debt sustainability and fiscal policy. *Revue économique*, 61(3), pp. 471–498.
- [12] Ostry, J.D., Ghosh, A.R., Kim, J.I. and Qureshi, M.S. (2010). Fiscal Space. *IMF Staff Position Note* SPN/10/11. Washington, D.C.: International Monetary Fund.
- [13] Reinhart, C.M. and Rogoff, K.S. (2010). Growth in a time of debt. *American Economic Review*, 100(2), pp. 573–578.
- [14] Reis, R. (2021). The constraint on public debt when $r < g$ but $g < m$. *American Economic Review: Insights*, 3(3), pp. 1–16.
- [15] Tokuoka, K. (2010). The Outlook for Financing Japan’s Public Debt. *IMF Working Paper* WP/10/19. Washington, D.C.: International Monetary Fund.
- [16] Willems, T. and Zettelmeyer, J. (2022). Sovereign Debt Sustainability and Central Bank Credibility. *Annual Review of Financial Economics*, 14(1), pp. 1–27.
- [17] Woodford, M. (1990). Public debt as private liquidity. *American Economic Review*, 80(2), pp. 382–388.